SUPCON

Introduction to Tier0

Last updated: 4th March 2026

Prepared by SUPCON International Business

Copyright © 2026 SUPCON. All rights reserved.

Challenges in Digital Transformation

SUPCON

Fragmented Data & System Landscape

- Disconnected IT / OT architectures
- Lack of enterprise-wide data governance
- Limited real-time operational visibility
- Redundant data ingestion & integration effort

Architecture Complexity

Unsustainable Cost Structure

- High Upfront Cost Per System
- Long Delivery, Delayed ROI Impact
- Scale Cost More than expected
- Low Reuse

Economic Non-Viability

Vendor Lock-in

- Platform dependency risk
- Limited interoperability & ecosystem integration
- Legacy technology stack constraints
- Restricted scalability across regions / plants

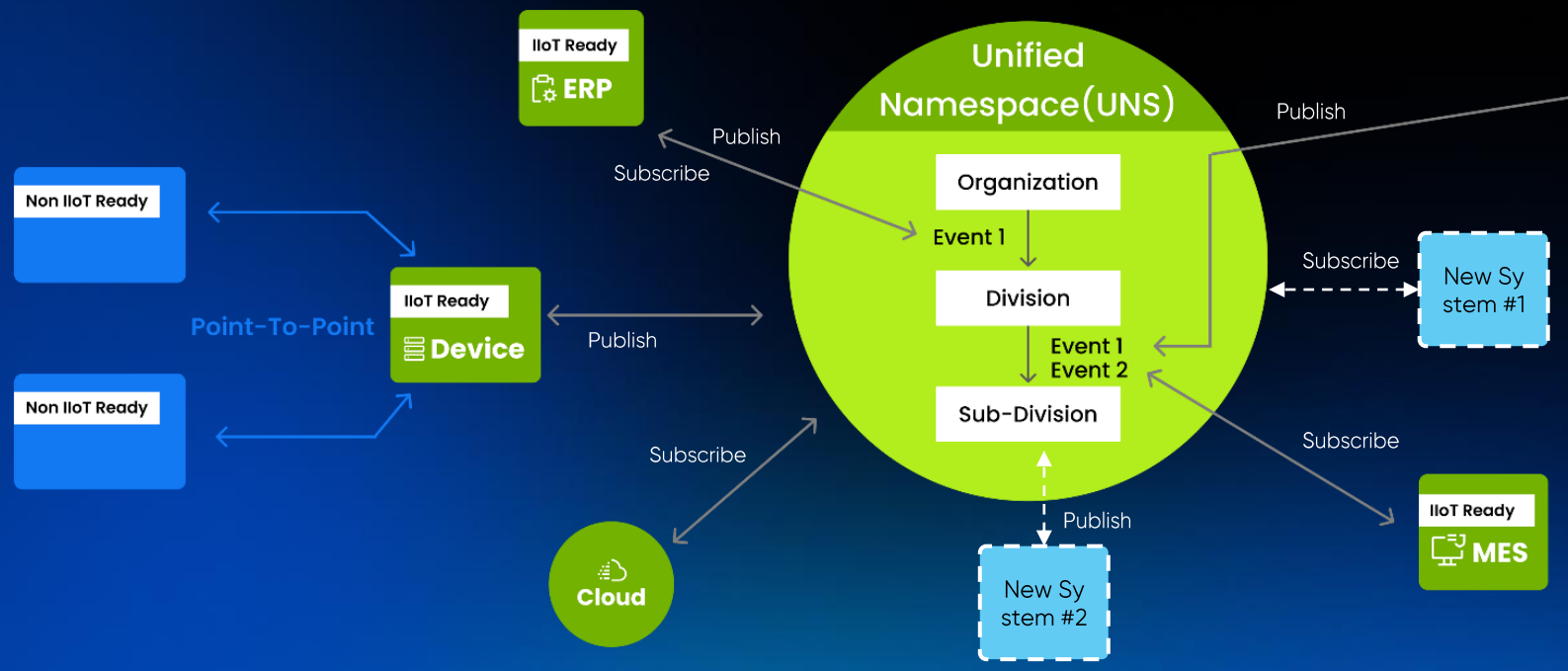
Strategic Dependency Risk

Copyright © 2026 SUPCON. All rights reserved.

UNS as the Data Foundation for Digitalization

SUPCON

Building the right data foundations is the first step to digitalization



Single Source of Truth

Any level of systems, equipment, and personnel within an enterprise can converse with a Single Source of Truth, achieving unparalleled data fluidity and interoperability.

Simple and Lightweight

Event-Driven Architecture

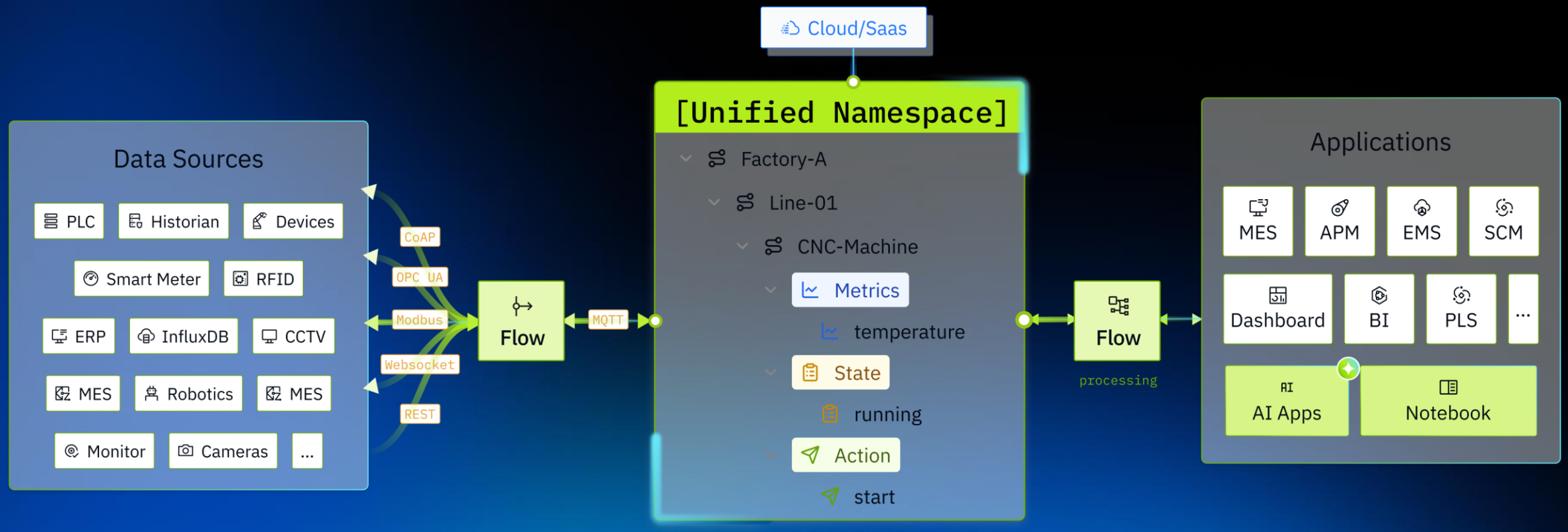
Highly Scalable

Real-time Snapshots

Copyright © 2026 SUPCON. All rights reserved.

Architecture

SUPCON



01 Open Structure

Built on open-source, industrial standards
Supports mainstream protocols

02 Faster delivery

Built-in end-to-end AI engine, significantly red ucing delivery time.

03 Future-Proof

AI ready data infrastructure

Copyright © 2026 SUPCON. All rights reserved.

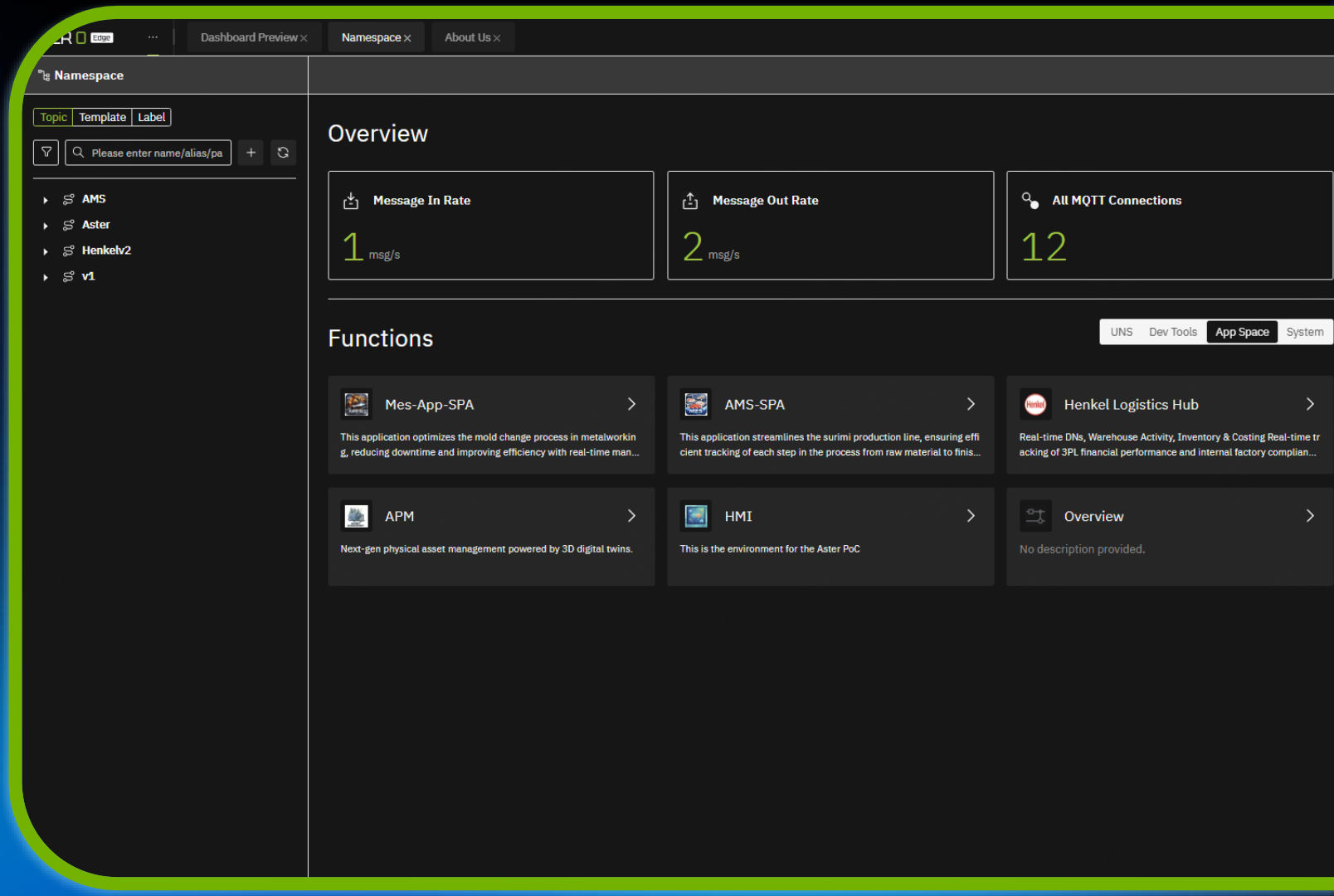
SUPCON

Tier0

The Best Open Source Practice For UNS

Github:
https://github.com/FREEZONEX/Tier0-Edge

Free trial:
https://tier0.app/trial



Copyright © 2026 SUPCON. All rights reserved.

Tier0 Open Technology Stack

SUPCON

Integration

KONGA

Kong

Visualizations

Grafana

AI Tools

marimo

Storage

Timescale

PostgreSQL

Message Broker

EMQ

Data Ingestion and Processing

Node-RED

Platform Foundations

docker

portainer.io

KEYCLOAK

elasticsearch

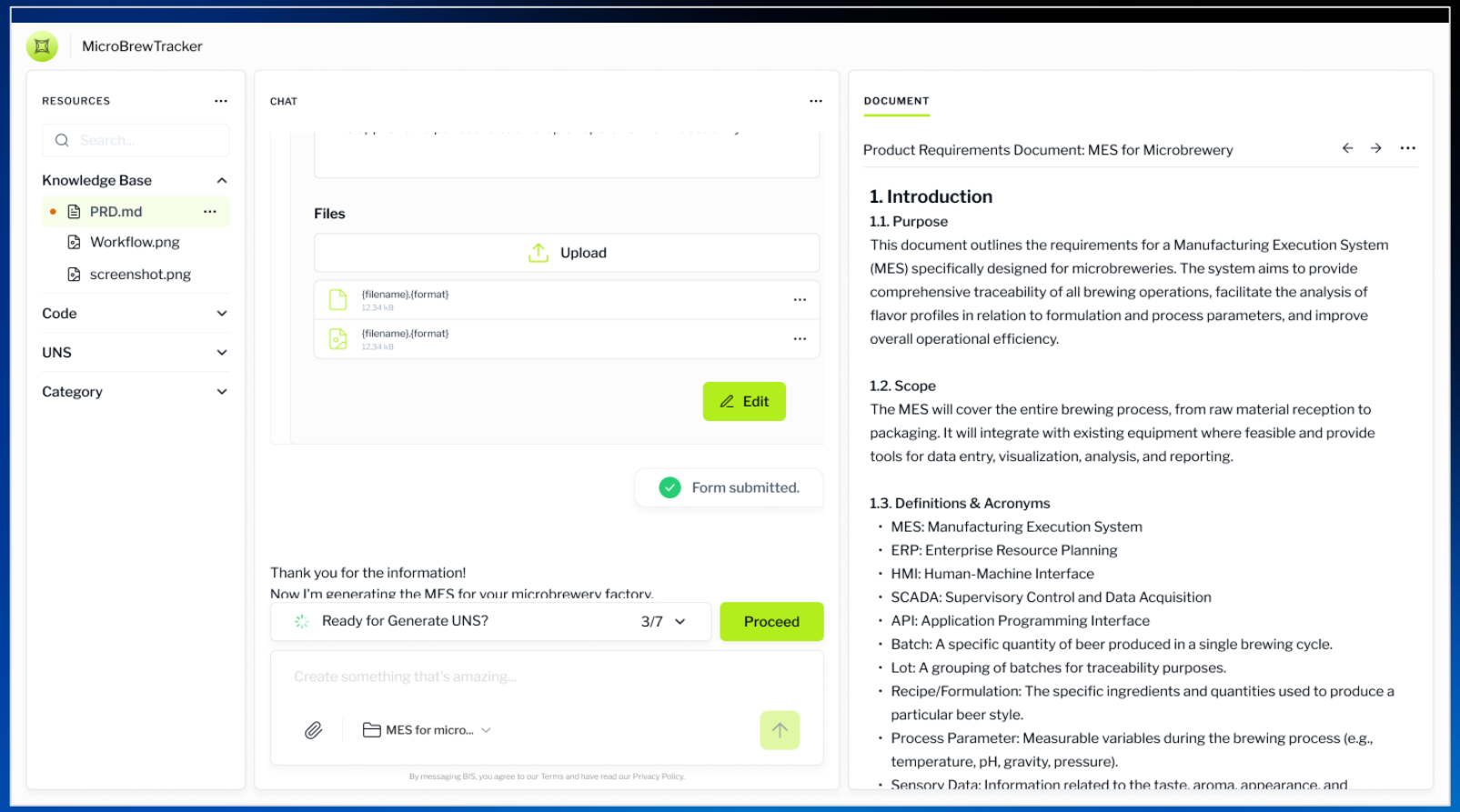
Gitea

Copyright © 2026 SUPCON. All rights reserved.

Embrace AI Capabilities

SUPCON

Generative Industrial APP Builder



- Add Context, More than just chat
- Add PDF, Drawing, Document

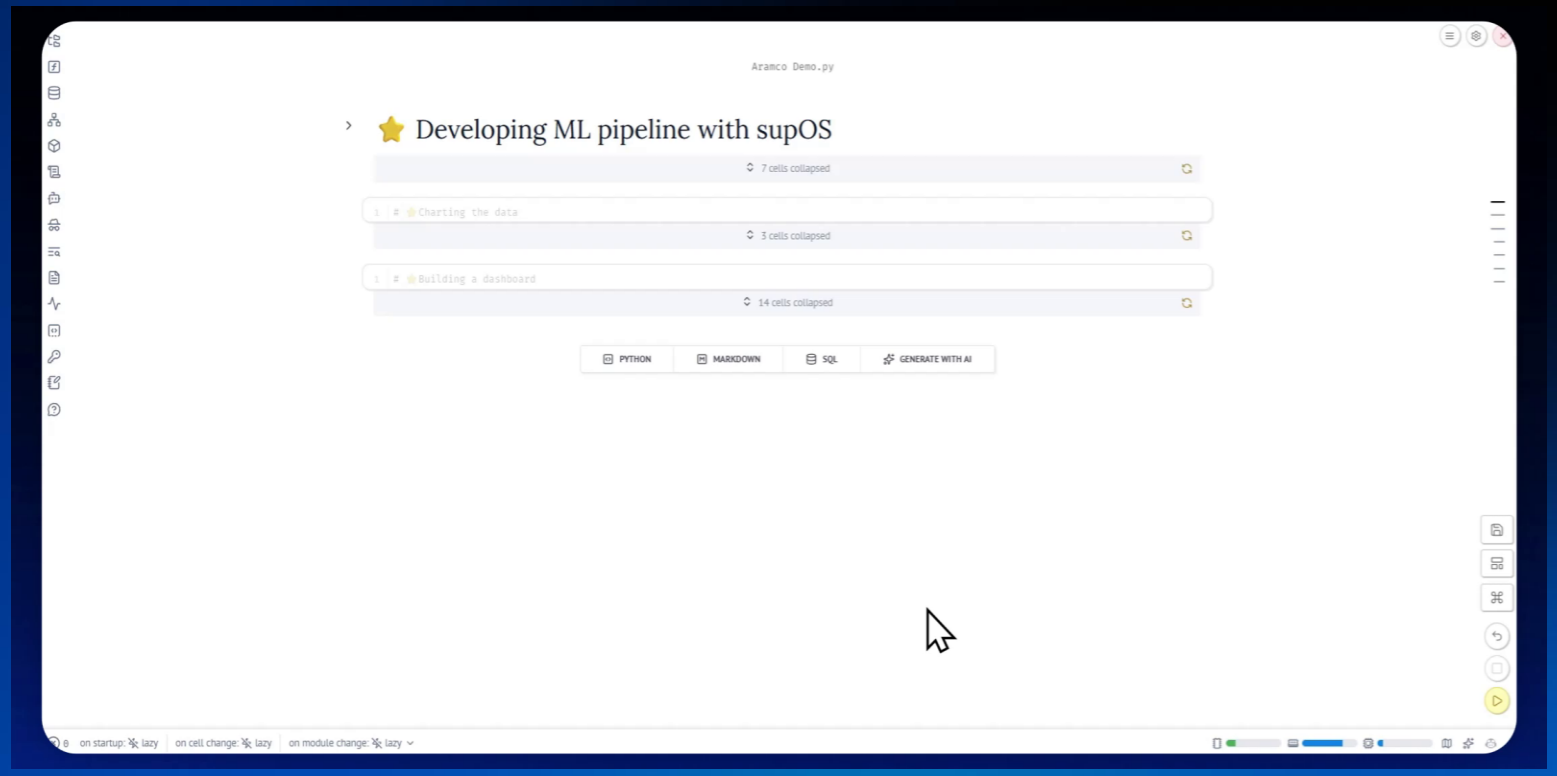
Let AI Design the APP

Copyright © 2026 SUPCON. All rights reserved.

Embrace AI Capabilities

SUPCON

Built-in Data Pipeline



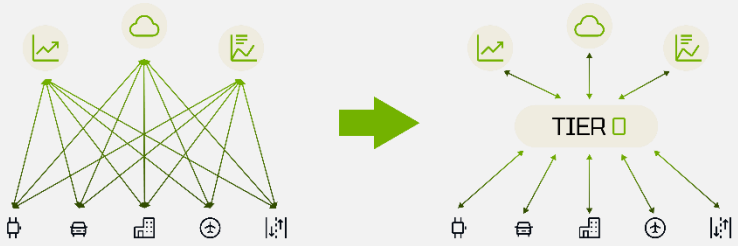
Built-in Marimo Notebook enables immediate pipeline design, data cleansing, and engineering workflows once connected to Tier 0.Your platform-level AI data assistant.

Copyright © 2026 SUPCON. All rights reserved.

Value Propositions of Tier0

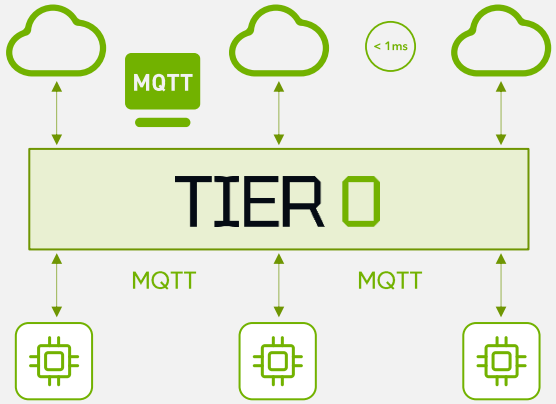
SUPCON

RESTful API vs MQTT



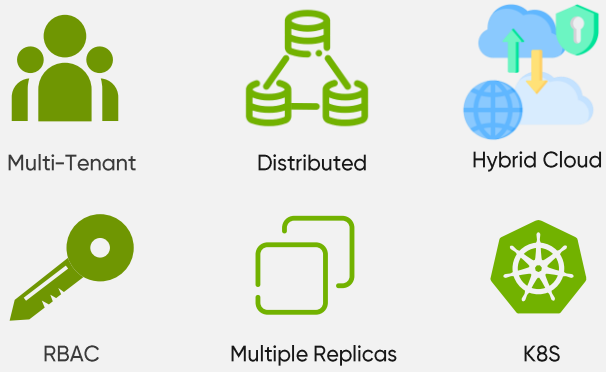
- Breaking the Data Spaghetti Dilemma
- Much easier to Scale Up
- Single-Source-of-Truth

Mission-Critical Data Fluency



- Large Amount of Data Ingestion
- Highly-Frequent Writing
- Time-Series ML Scenarios
- High Volume of Data Distribution

Modern & Flexible Stack



- Flexible Cloud/Edge/Hybrid Deployment
- Component-level HA, Multiple Copies
- Distributed, Empower Easy Scaling Busin ess
- Web and Cross Platform

Copyright © 2026 SUPCON. All rights reserved.

Product Editions & Pricing

SUPCON

Open Source

$0 / year

Free

Get Started

- Core Features Open-sourced
- Free Trial
- Regular Technical Support

Enterprise

Starts From

$29,000 / year

Annual Subscription

Get Started

- High Availability
- Private Deployment Support
- Premium Service Available

SaaS

Starts From

$25,000 / year

Annual Subscription

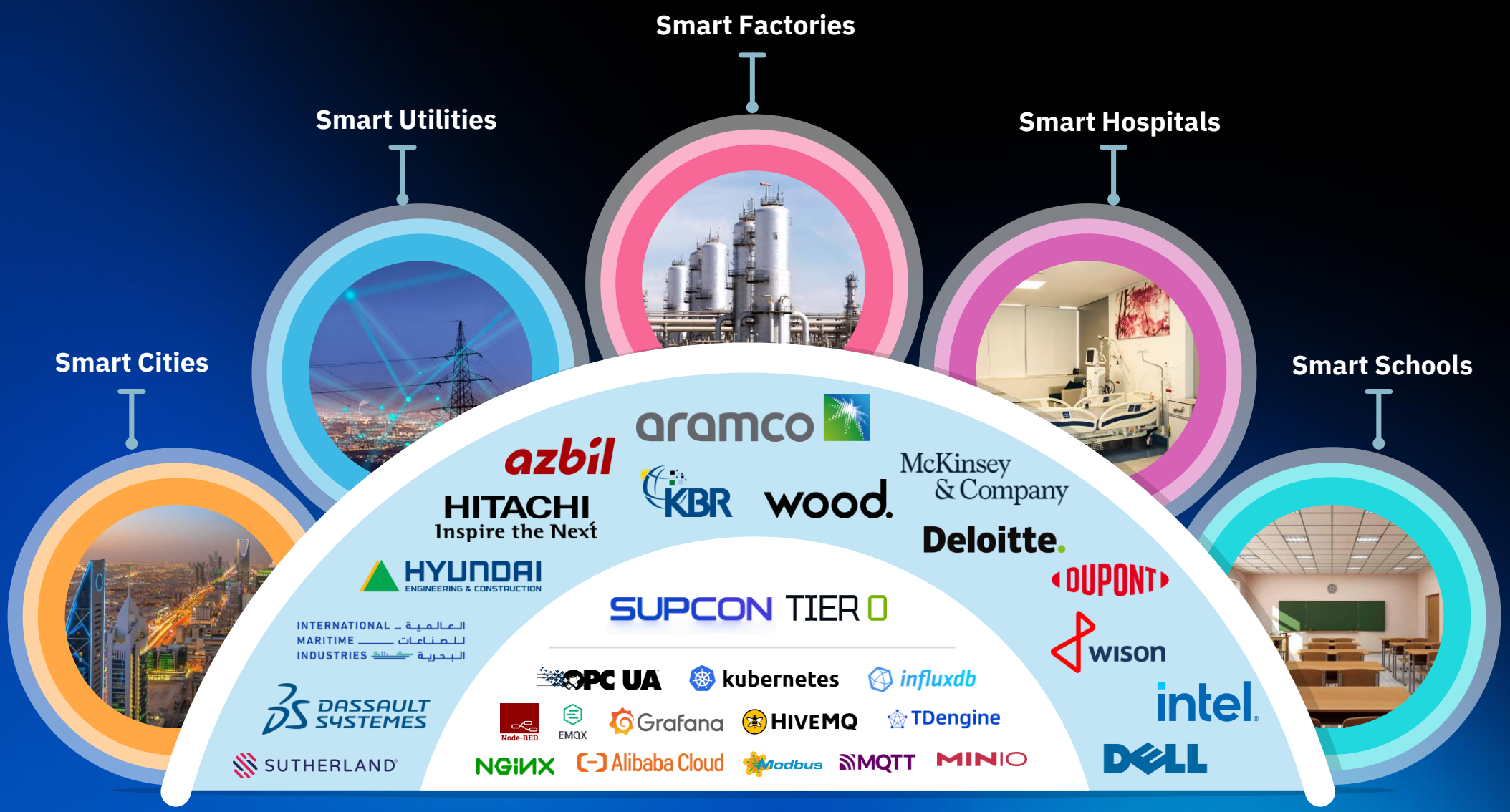
Coming Soon

- Public Cloud On AWS
- Fully Managed Service
- Premium Service Available

Copyright © 2026 SUPCON. All rights reserved.

Drive Customer Success

SUPCON



Copyright © 2026 SUPCON. All rights reserved.

Case Study
Aramco City Brain Project

SUPCON

Background

This project delivered a next-generation Building Management System (BMS) for a building within Aramco's Dammam campus, leveraging the "City Brain" architecture to establish an open, scalable, and future-ready smart building platform.

Key Benefits

AGILE START	SCALABILITY	TRUST
Low	**Modular**	**Open**
ENTRY COST	COMPOSABLE STACK	BY DESIGN
Scenario-First	Expand As Grow	Secure Platform

Deliverables

Copyright © 2026 SUPCON. All rights reserved.

Case Study

Sugar Mill Plant Real-Time Performance Dashboard

SUPCON

Background

- Sugar mill operations span multiple plants and processes, with data scattered across different systems. Limited real-time visibility and delayed reporting made it difficult to optimize production, energy usage, and operational efficiency.

Highlight

- Real-time visibility across multiple mills
- Centralized data integration & KPI monitoring
- Production and energy performance analytics
- Predictive insights to reduce bottlenecks
- Executive-ready dashboards and reports

Copyright © 2026 SUPCON. All rights reserved.

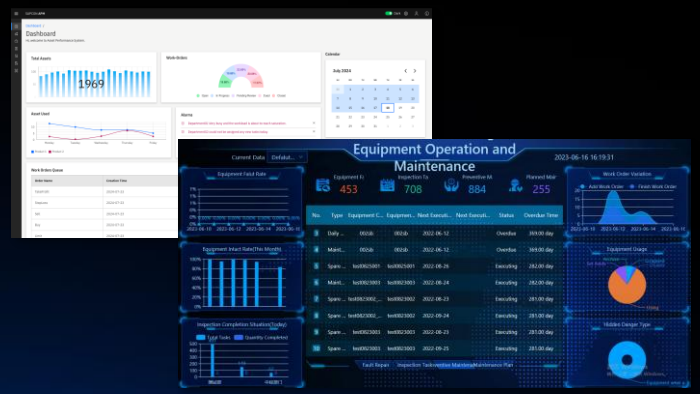
Examples of Use Cases Enabled by Tier0

SUPCON

Production Management
Real-Time Production Monitoring, OEE/KPI Dashboards

Safety Management / EHS
HAZOP, PLS, Video Surveillance, Alarm Systems, Notifications

Asset Management
E2E Lifecycle, Predictive Maintenance, Equipment Health



Supply Chain Management
Warehouse Visibility, Traceability, Lean Operations

Energy Management
Energy Monitoring, Analysis and Optimization

Quality Management / LIMS
Quality Monitoring, Process Optimization

Copyright © 2026 SUPCON. All rights reserved.

SUPCON
THANK YOU
在世界工业文明的进程中
留下深深的足迹
跑起来就有风
2023
智能趋势论坛